

Design Verification Plan & Report (virtual test) — VBF-T6R1-DVPR-01

Candidate: Inmo_r8_d6 (bridge/params_final.json). All results DFN fidelity, tool-output sourced. One row per verification item.

#	Item		Condition	Result value		Determination	Source
1	1C discharge capacity		1C, 25 °C ambient	6.198 Ah		informational (no capacity criterion in contract)	cell/r8_d6_1c_dfn.json:capacity_ah
2	Volumetric energy density	1C discharge energy ÷ cell volume	1135.8 Wh/L		✓ ≥ 950 Wh/L (+19.6%)	cell/r8_d6_energy_dfn.json:energy_density_wh_l	
3	Voltage plateau		discharge midpoint	4.1145 V		✓ ≥ 4.1 V (+14.5 mV)	cell/r8_d6_energy_dfn.json:midpoint_voltage_v
4	4C fast-charge temperature rise		4C CC charge to 4.7 V at 45 °C ambient (protocol: 1C discharge to 2.5 V + 4C charge), lumped thermal	321.42 K = 48.27 °C		✓ ≤ 323.15 K (margin 1.73 K)	cell/r8_d6_4c_dfn.json:T_max_K
5	4C fast-charge plating	min anode surface potential over full protocol	+0.01096 V		✓ plated = false (margin 11.0 mV; min occurs at charge end)	cell/r8_d6_4c_dfn.json:anode_potential_v	
6	SEI thickness after 100 cycles	aging_1C_100cyc, DFN, isothermal	385.3 nm		✓ ≤ 500 nm (margin 114.7 nm)	cell/r8_d6_aging_dfn.json:sei_thickness_nm_end	
7	DC resistance		midpoint DCR	6.38 mΩ		informational	cell/r8_d6_energy_dfn.json:dcr_ohm

Items explicitly N/A (beyond pure simulation boundary, require physical experiment): nail penetration; overcharge to thermal runaway; crush; drop; cycle life to end-of-life (only 100-cycle SEI growth simulated — no capacity-fade EOL model); rate-pulse internal resistance (DCR reported is midpoint-calculated, not pulse).

Conclusion: All five contract criteria PASS at DFN fidelity (verified items 2–6). Uncovered conditions listed above are cited here directly as paper limitations. Verification is virtual — no physical cells were built or tested.